

PLYMOUTH, NH, USA

WMO: 742078

Lat: 43.779N

Lon: 71.754W

Elev: 154

StdP: 99.49

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 64773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.4	(c) -17.8	(d) -26.1	(e) 0.4	(f) -18.9	(g) -23.6	(h) 0.5	(i) -15.6	(j) 9.8	(k) -6.9	(l) 9.0	(m) -7.6	(n) 2.2	(o) 280	(p) 0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	30.9	21.1	29.1	20.4	27.7	19.5	22.9	27.9	22.0	26.9	21.1	25.4	2.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.3	25.1	20.5	15.4	24.2	19.7	14.7	23.3	68.6	28.1	65.2	27.1	61.7	25.6	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.3	(c) 6.0	(e) -25.8	(f) 33.5	(g) 3.2	(h) 1.0	(i) -28.2	(j) 34.2	(k) -30.0	(l) 34.8	(m) -31.8	(n) 35.3	(o) -34.2	(p) 36.0		
(a) 8.3	(b) 7.3	(c) 6.0	(e) -26.0	(f) 24.6	(g) 3.0	(h) 0.9	(i) -28.2	(j) 25.3	(k) -29.9	(l) 25.8	(m) -31.6	(n) 26.4	(o) -33.9	(p) 27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-6.4	-5.2	-0.5	6.3	13.2	17.4	19.3	15.4	9.4	2.5	-2.9	(6)	
(7)		DBStd	10.37	5.79	5.55	5.37	4.50	4.16	3.42	2.83	2.72	4.16	4.23	4.39	5.09	(7)
(8)		HDD10.0	2102	509	425	328	128	14	0	0	4	64	229	401	(8)	
(9)		HDD18.3	4150	768	658	584	360	168	57	13	21	107	278	476	659	(9)
(10)		CDD10.0	1176	0	0	2	18	113	222	320	288	166	44	3	0	(10)
(11)		CDD18.3	183	0	0	0	1	8	29	74	51	19	1	0	0	(11)
(12)		CDH23.3	1993	0	0	4	22	182	358	751	471	194	11	0	0	(12)
(13)		CDH26.7	541	0	0	0	6	51	99	229	104	51	1	0	0	(13)
(14)	Wind	WSAvg	1.7	2.3	2.4	2.2	1.7	1.4	1.1	1.0	1.1	1.5	1.7	2.1	(14)	
(15)	Precipitation	PrecAvg	1111	75	67	89	86	87	102	113	99	117	94	96	(15)	
(16)		PrecMax	1388	148	165	135	164	206	224	200	240	218	274	140	177	(16)
(17)		PrecMin	777	18	27	42	15	12	33	43	17	37	27	52	40	(17)
(18)		PrecStd	163	32	28	26	38	46	51	39	51	38	71	22	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	11.8	20.2	26.4	31.1	32.4	33.2	31.4	31.1	25.1	17.9	10.8	(19)
(20)			MCWB	4.5	6.0	12.9	15.3	19.9	21.1	23.1	22.2	22.0	16.8	13.4	9.1	(20)
(21)		2%	DB	5.0	7.3	13.0	22.2	27.8	29.6	31.0	29.1	28.0	21.3	14.8	7.4	(21)
(22)			MCWB	2.6	3.4	6.6	12.7	17.5	19.9	21.6	20.9	20.5	15.5	10.8	5.2	(22)
(23)		5%	DB	2.8	3.9	9.3	18.7	25.1	27.2	29.2	27.7	25.5	18.4	12.1	5.5	(23)
(24)			MCWB	1.0	1.4	5.0	10.9	16.2	18.2	20.9	20.1	18.9	14.6	9.3	3.6	(24)
(25)		10%	DB	1.4	2.0	6.7	15.3	22.3	25.2	27.5	26.2	22.9	16.2	9.6	3.4	(25)
(26)			MCWB	0.1	0.1	3.0	8.7	14.4	17.3	20.0	19.5	17.8	12.9	7.1	1.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.3	13.3	17.2	21.4	22.7	24.8	23.5	23.4	19.5	14.8	8.6	(27)
(28)			MCDWB	7.5	10.2	19.6	23.1	27.0	28.0	30.3	28.8	28.5	21.8	16.8	9.5	(28)
(29)		2%	WB	2.8	3.6	8.4	14.7	19.6	21.6	23.4	22.3	21.7	16.8	12.0	5.8	(29)
(30)			MCDWB	4.3	6.1	11.4	21.0	25.1	27.2	28.4	27.2	26.6	19.6	13.8	6.8	(30)
(31)		5%	WB	1.5	1.8	5.3	11.8	17.8	20.2	22.3	21.3	20.3	15.2	9.4	3.8	(31)
(32)			MCDWB	2.6	3.7	7.8	16.4	22.2	24.5	26.9	25.7	23.8	17.5	11.6	5.4	(32)
(33)		10%	WB	0.3	0.5	3.6	9.9	16.2	18.9	21.3	20.5	18.9	13.7	7.5	2.2	(33)
(34)			MCDWB	1.2	1.6	6.3	14.9	20.3	23.0	25.5	24.3	21.9	16.0	9.0	3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.3	10.4	11.6	13.3	12.2	12.5	12.7	10.8	9.2	7.7	(35)	
(36)			MCDBR	9.5	12.0	14.9	19.4	18.5	16.1	15.4	14.7	15.1	14.3	13.8	9.2	(36)
(37)			MCWBR	7.8	8.8	9.4	10.3	9.1	7.4	7.3	7.4	8.1	8.9	9.9	7.7	(37)
(38)		5% WB	MCDWR	8.6	10.9	13.0	15.9	14.5	13.0	12.8	12.8	13.0	11.6	11.9	8.6	(38)
(39)			MCWBR	7.4	8.4	9.0	9.8	7.8	7.0	6.7	6.8	7.6	8.2	9.6	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.287	0.316	0.352	0.380	0.428	0.445	0.417	0.370	0.335	0.309	0.281	(40)	
(41)		taud	2.404	2.345	2.389	2.402	2.373	2.248	2.212	2.277	2.395	2.504	2.519	2.472	(41)	
(42)		Ebn at Noon	862	910	925	913	892	846	825	837	855	847	817	822	(42)	
(43)		Edh at Noon	75	94	104	112	119	135	139	126	103	81	68	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.59	2.46	3.58	4.45	4.99	5.40	5.59	5.07	4.12	2.54	1.68	1.23	(44)	
(45)		RadStd	0.14	0.22	0.23	0.52	0.59	0.49	0.40	0.28	0.27	0.26	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78	N/A

Nomenclature: See separate page